

Where Query-Conditioned Audio Ranking Transfers—and Where It Fails: Controlled Confirmation, Frozen Natural Evaluation, and Cap-Corrected Diagnosis

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Abstract—Exact-onset mixtures make temporal evidence ranking auditable, but may not predict localization in natural recordings. We evaluate a five-seed query-conditioned residual (QCR) ranker that perturbs a pinned CLAP cosine prior by at most 0.25. On a frozen source-isolated panel, evidence average precision rises from 0.6090 to 0.8425; an external source-disjoint panel retains a positive 0.1612 gain but misses its strict absolute-performance gate. We then score all 5,359 Perception Test validation videos and 35,625 events under a frozen zero-shot protocol. Multiscale QCR obtains 0.052603 mAP versus 0.067576 for CLAP, with every temporal-IoU threshold worse and only two of six gates passing. A disclosed post-outcome, five-fold diagnostic selects scorer and temporal scale per class, then reapplies one shared 200-segment video cap. It reconstructs all ten capped candidates exactly and reaches 0.096995 mAP. However, a CLAP-only router reaches 0.093921; QCR contributes only 0.3074 percentage points, or 10.45% of the gain over fixed CLAP. The full router remains below supervised ActionFormer and the learned retrieval fails a held-out answer bridge. The combined evidence confirms QCR for controlled evidence ranking, shows that fixed QCR fails zero-shot transfer to natural audio, and identifies class-dependent temporal scale as the dominant in-domain mechanism. The cross-fitted result is diagnostic, not fresh confirmation.

Index Terms—audio event grounding, temporal sound localization, CLAP, transfer evaluation, cross-fitting, temporal scale, reproducibility

I. INTRODUCTION

TEXT-CONDITIONED audio retrieval is frequently studied by inserting a known sound into a longer background. Exact insertion times make this design useful: every chunk has a precise overlap label, distractors can be controlled, and source leakage can be audited. Natural recordings are harder. Events overlap, vary in duration and loudness, admit ambiguous class descriptions, and compete under a finite proposal budget. A ranker can therefore excel at controlled evidence retrieval without transferring to natural temporal localization.

Contrastive language–audio pretraining (CLAP) provides a transparent zero-shot similarity between an audio segment and text [1], [2]. A bounded learned residual may refine this prior, but positive controlled results leave at least three questions. Does the reordering survive a change of acoustic construction? Does it survive natural polyphony when temporal scale and output competition are held fixed? If localization improves, does that evidence help an audio-language model answer questions?

We address these questions as an evidence ladder rather than forcing one headline across incompatible stages. The same five frozen QCR checkpoints first enter a source-isolated exact-onset confirmation and then a source-disjoint external construction. A one-shot natural transfer protocol evaluates all 5,359 Perception Test sound-localization validation videos [12]. It freezes prompts, scales, post-processing, caps, class-macro mean average precision (mAP), a video-paired bootstrap, and six gates before natural scores exist.

The frozen natural result is negative: multiscale QCR reduces mAP relative to multiscale CLAP, with all five threshold differences adverse. We retain this boundary. We then ask a different, explicitly post-outcome question: can a low-capacity in-domain router diagnose whether scorer or temporal scale is responsible? Ten CLAP/QCR candidates span 0.5, 1, 2, 4, and pooled scales. For each of five video-hash folds and ten classes, the candidate is chosen using only the other four folds. Selected class streams are merged on the held fold and subjected to one shared 200-segment video cap. A CLAP-only version isolates scale selection from the residual ranker.

Our contributions are:

- 1) a multi-stage, outcome-faithful evaluation of a bounded query-conditioned ranker, including positive controlled evidence and a retained negative natural-transfer decision;
- 2) a cap-corrected cross-fitting protocol whose candidate pools reproduce all original capped outputs exactly before class routing;
- 3) a mechanism decomposition showing that class-dependent temporal scale accounts for nearly 90% of the exploratory gain over fixed multiscale CLAP, while QCR supplies a smaller positive increment; and
- 4) a downstream answer boundary, supervised contextual reference, source-linked figures, immutable prediction hashes, protocol invalidations, and executable verification.

The post-outcome router is not described as zero-shot or independent confirmation. Its role is to turn a failed transfer result into a testable mechanistic diagnosis.

Evidence ladder: positive controlled ranking, failed natural transfer, and post-outcome diagnosis

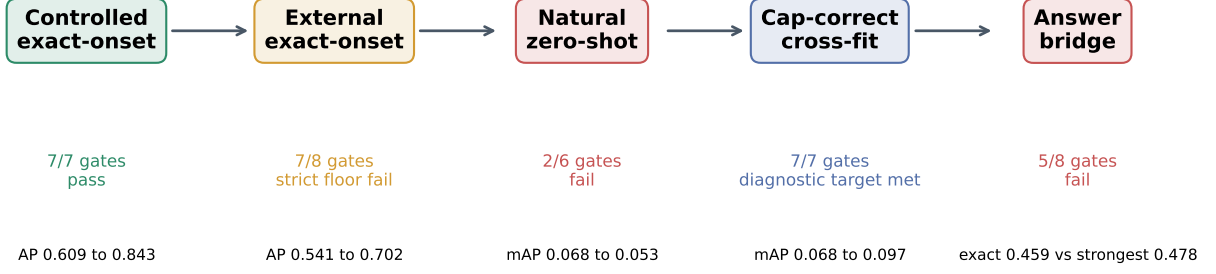


Fig. 1. Evidence ladder and immutable decisions. Controlled and natural endpoints use frozen gate conjunctions. Cross-fitting uses the already exposed natural validation panel and supports diagnosis only.

II. RELATED WORK

A. Language–audio representation and grounding

CLAP aligns audio and text through contrastive pretraining [1]; the LAION implementation expands data and studies feature fusion and keyword-to-caption augmentation [2]. AudioCLIP learns a joint image–text–audio space [3], while Wav2CLIP distills audio into CLIP’s visual-language space [4]. These encoders offer auditable segment scores but are not trained directly for temporal boundaries.

Text-to-audio grounding links phrases to event times [5]. Later work studies cross-modal interaction, suppression of irrelevant regions, and weak caption supervision [6], [7]. Auto-AEG combines synthetic exact intervals with pseudo-labeled real audio for scalable open-vocabulary grounding [8]. Our study is narrower: queries are renderings of ten known benchmark classes, and we test whether a ranker developed on controlled mixtures transfers without natural-label fitting.

B. Sound-event detection and temporal localization

DESED studies how non-target events, reverberation, label granularity, and synthetic-versus-recorded data affect event detection [9]. PSDS reduces dependence on one operating point and exposes class stability [10]. ActionFormer uses multiscale temporal features and boundary regression for proposal-free localization [11]. Perception Test adapts temporal localization to diagnostic multimodal video and audio [12]; TIM represents audio/visual time intervals explicitly [13]. These results motivate our separation of scale, scorer, class, and supervision.

C. Cross-fitting and benchmark exposure

Cross-fitting prevents each held-fold outcome from choosing its own candidate, but it does not make a reused public validation set untouched. Our folds estimate a low-capacity in-domain routing rule. Candidate families were considered after the frozen natural failure, so the result remains post-outcome exploratory. This distinction is more important than the word “held-out”: fold isolation controls local selection leakage, whereas fresh external confirmation requires an unexposed panel.

III. QUERY-CONDITIONED RANKING

A. CLAP prior and bounded residual

For audio window i and class query q , the pinned unfused CLAP encoder produces normalized embeddings $\mathbf{a}_i, \mathbf{q} \in \mathbb{R}^{512}$ and prior

$$p_i = \mathbf{a}_i^T \mathbf{q}. \quad (1)$$

Three templates per class are embedded and averaged after normalization. The QCR ranker projects audio and query embeddings to 64 dimensions; features include both projections, product, absolute difference, p_i , normalized position, and a sinusoidal position term. Its score is

$$s_i = p_i + 0.25 \tanh(g_i), \quad (2)$$

so $|s_i - p_i| \leq 0.25$ for every proposal. Five checkpoints trained with seeds 0–4 are averaged without seed selection on later panels. Each has 82,759 trainable parameters.

B. Controlled exact-onset task

A target sound is mixed with backgrounds and distractors. Four-second chunks at two-second hop receive overlap-derived evidence labels. Evidence AP is the primary ranking endpoint; exact-onset Hit@1, Recall@4, and top-chunk IoU are secondary. The source-isolated confirmation contains 397 recipes, 34,193 chunks, and 2,956 unique source files with zero development-recipe or source overlap. Uncertainty resamples recipes.

The external construction uses UrbanSound8K target clips after excluding Freesound source IDs present in ESC-50, with fresh LibriSpeech test-other backgrounds. It contains 219 recipes and resamples original UrbanSound source IDs. Its strict conjunction additionally requires an absolute performance index $50(\text{AP} + \text{Hit@1}) \geq 85$.

IV. NATURAL-AUDIO EVALUATION

A. Frozen zero-shot protocol

The natural panel contains all 5,359 Perception Test validation recordings, 35,625 annotated events, and ten sound

classes. No Perception Test training label fits QCR. Annotation metadata was inspected to choose windows (w, h) in seconds:

$$(0.5, 0.25), (1, 0.5), (2, 1), (4, 2).$$

CLAP and QCR share proposals, prompt ensemble, sigmoid score transform, Gaussian soft-NMS [14], at most 50 segments per class, and one 200-segment video cap. The four frozen methods are CLAP/QCR at four seconds and CLAP/QCR with pooled scales.

For class c and temporal-IoU threshold τ , predictions are sorted by confidence and matched one-to-one to ground-truth intervals. The endpoint is

$$\text{mAP} = \frac{1}{5} \sum_{\tau \in \{0.1, \dots, 0.5\}} \frac{1}{10} \sum_c \text{AP}_{c, \tau}. \quad (3)$$

The paired uncertainty summary computes macro AP within each video and resamples all videos 10,000 times. It is complementary to global class-macro mAP and can have a different numerical effect size.

Promotion requires all six conditions: positive global difference, positive video-bootstrap lower endpoint, improvement at at least four thresholds, no class loss worse than -0.05 , QCR multiscale at least as good as CLAP at four seconds, and complete integrity. Prompts, scales, seeds, checkpoints, metric, and gates cannot change after scoring.

B. Cap-corrected cross-fitted diagnosis

After the frozen failure, six additional single-scale candidate outputs were generated, giving CLAP and QCR at 0.5, 1, 2, 4, and pooled scales. The first exploratory router was invalidated before a report because joining separately capped class outputs could bypass the shared 200-segment video limit. The corrected track rebuilds per-class pools before that shared cap and verifies that reapplying the original cap reconstructs all ten source candidate files row for row.

A SHA-256 video hash defines five folds. For held fold f and class c , the candidate with highest class AP on the other four folds is selected, with a stable name tie break. On fold f , only that candidate's class- c proposals are retained. All selected classes are then merged, sorted, and capped once at 200 segments per video. The CLAP-only ablation repeats the procedure with QCR candidates excluded.

This protocol gives 50 fold-class selections. It uses natural class labels and an already exposed validation benchmark. Its criteria test internal coherence—positive global and paired effects, threshold breadth, class non-harm, cap reconstruction, complete assignment, and integrity—not zero-shot promotion.

V. RESULTS

A. Controlled and external exact-onset evidence

On the source-isolated confirmation, CLAP evidence AP is 0.6090 and QCR AP is 0.8425, a gain of 0.2335 with 95% recipe-bootstrap interval $[0.1850, 0.2841]$. Hit@1 rises from 0.7683 to 0.9446. All five seeds improve and all seven gates pass.

On the external source-disjoint construction, AP rises from 0.5408 to 0.7020; the 0.1612 gain has 95% source-bootstrap

interval $[0.0911, 0.2334]$. Hit@1 rises from 0.6256 to 0.7534. Seven of eight gates pass. The declared performance index is 72.77, below 85, so the strict external promotion decision remains a failure despite replicated relative gains.

B. Frozen natural transfer failure

Multiscale CLAP obtains 0.067576 mAP; multiscale QCR obtains 0.052603. The frozen QCR-minus-CLAP difference is -1.4973 percentage points. All five temporal-IoU differences are negative. The paired-video mean change is -1.1334 points, with interval $[-1.4749, -0.7804]$ and zero of 10,000 replicates above zero. The worst class change is -7.1301 points. Only the scale-floor and integrity gates pass.

The official supervised audio-only ActionFormer checkpoint obtains 0.116638 mAP under the same recomputed metric. It uses Perception Test training labels and official features, so it is context rather than a zero-shot gate.

C. What cap-corrected cross-fitting diagnoses

The full cap-corrected router obtains 0.096995 mAP versus 0.067576 for fixed multiscale CLAP, a gain of 2.9419 points (43.53% relative). Every threshold and class difference is positive. The paired-video macro-AP effect is 6.2251 points with interval $[5.8165, 6.6397]$. All seven diagnostic criteria pass.

The CLAP-only router already obtains 0.093921, a gain of 2.6345 points over fixed CLAP. Adding QCR raises global mAP by only 0.3074 points, 10.45% of the total gain. Its post-hoc paired-video increment is 1.3737 points with interval $[1.1571, 1.5895]$. Thus QCR has a positive in-domain increment, but class-dependent scale choice is the dominant mechanism.

QCR is selected for 30.0% of the 50 fold-class assignments. Nine of ten classes choose the same candidate in all folds. Speech and animal sound prefer QCR at four seconds; dropping prefers QCR at one second. Clapping selects half-second CLAP, fluid selects two-second CLAP, and the remaining classes choose one-, two-, or four-second CLAP. This stability explains why a pooled-scale score is not optimal under cross-class proposal competition.

The full route merges 1,453,945 proposals, discards 409,789 under the shared limit, and writes 1,044,156 predictions; 4,401 videos exceed 200 proposals before capping. The CLAP-only route is audited identically. All ten candidate reconstructions are exact, no output video exceeds 200 segments, and no metric is nonfinite. The full router remains 1.9643 points below supervised ActionFormer.

D. Downstream answer boundary

On 411 held-out answer examples from 25 sources and two audio-language decoders, learned QCR retrieval obtains two-backbone source-macro exact accuracy 0.4586. CLAP retrieval obtains 0.4619, deterministic random retrieval 0.4731, the strongest eligible prefix-matched baseline 0.4780, and oracle spans 0.4886. The learned method passes five of eight gates and does not improve answer accuracy. This result is

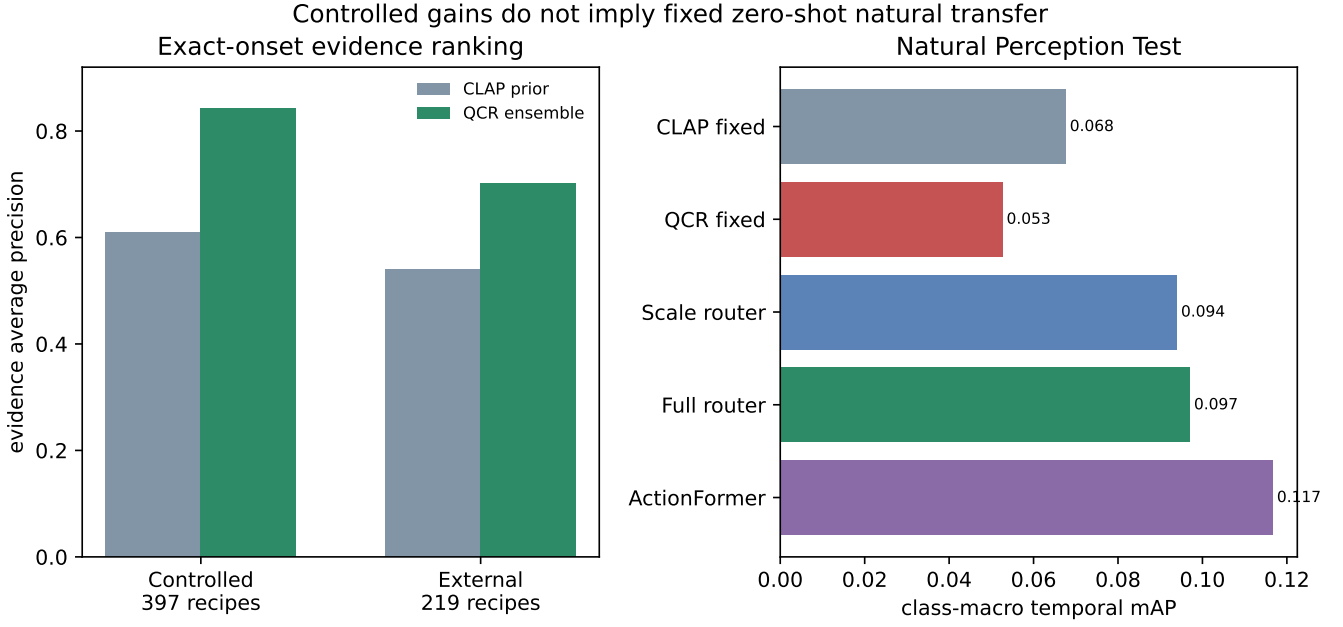


Fig. 2. Left: QCR improves exact-onset evidence AP on both controlled constructions. Right: fixed QCR degrades frozen natural mAP; post-outcome cross-fitting recovers much of the supervised-context gap. Bars on the right have different evidence status and are not one leaderboard.

TABLE I
EVIDENCE-STAGE LEDGER. “PASS” ALWAYS REFERS TO THAT STAGE’S DECLARED CONJUNCTION, NOT TO A UNIVERSAL MODEL CLAIM.

Stage	Evidence status	Panel	Primary contrast	Gates / decision
Controlled exact-onset	Frozen confirmatory	397 recipes	AP +0.2335	7/7 pass
External exact-onset	Frozen external	219 recipes	AP +0.1612	7/8; strict floor fail
Natural transfer	Frozen zero-shot	5,359 videos	mAP −1.4973 pp	2/6 fail
Cap-correct router	Post-outcome cross-fit	same validation	mAP +2.9419 pp	7/7 diagnostic criteria
Answer bridge	Held-out downstream	411 examples	exact 0.4586 vs. 0.4780	5/8 fail

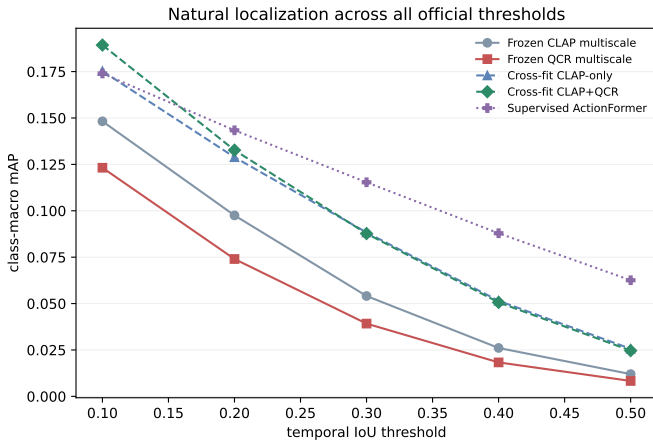


Fig. 3. Natural class-macro mAP at every official tIoU threshold. Fixed QCR is below fixed CLAP throughout. Cross-fitted curves are post-outcome; the ActionFormer curve is supervised context.

compatible with the localization analysis: overlap AP does not ensure that a fixed-duration span contains the evidence a decoder needs.

TABLE II
NATURAL LOCALIZATION STATUS AND DECOMPOSITION.

Method	mAP	Evidence role
Fixed CLAP multiscale	0.067576	frozen zero-shot comparator
Fixed QCR multiscale	0.052603	frozen zero-shot primary
Cross-fit CLAP-only	0.093921	post-outcome scale diagnosis
Cross-fit CLAP+QCR	0.096995	post-outcome full diagnosis
ActionFormer audio	0.116638	supervised context

VI. DISCUSSION

A. A transfer boundary, not a contradictory result

Controlled and natural tasks differ in both data and relevance structure. In exact-onset mixtures, one inserted target defines positive chunks and distractors are engineered. In natural audio, several valid events can overlap, class durations differ, and all class proposals compete under a shared cap. QCR’s bounded residual can learn a useful controlled ordering and still harm the fixed natural ordering. The frozen negative result is therefore a direct measurement of construct transfer, not a failed implementation.

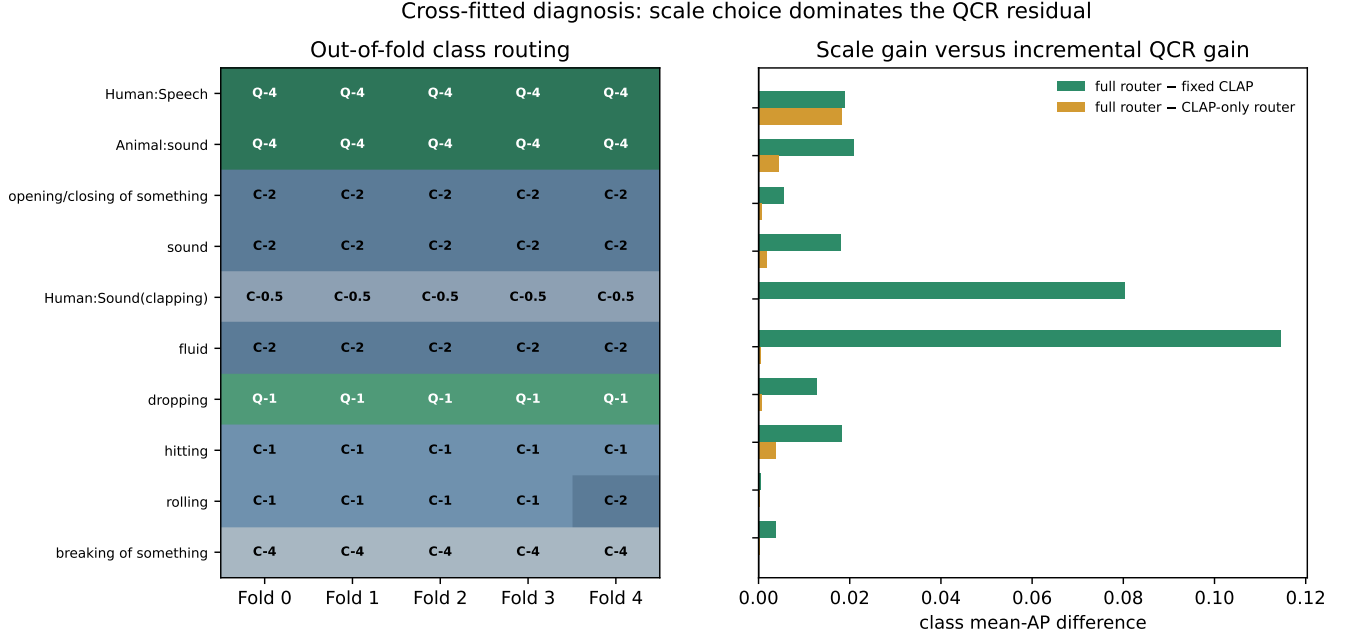


Fig. 4. Left: selected candidate for every held fold and class (C =CLAP, Q =QCR, number=window seconds). Nine classes are unanimous. Right: the full router’s gain over fixed CLAP is large for clapping and fluid; the incremental QCR gain beyond the CLAP-only router is smaller but positive for every class.

B. Why temporal scale dominates

Short impulses such as clapping benefit from half-second windows, while speech and broad object sounds prefer longer context. Pooling every scale does not merely add recall: it changes proposal density and soft-NMS/cap competition. Per-class selection recovers 2.63 mAP points with CLAP alone. QCR adds 0.31 points globally and a positive video-paired increment, mainly where longer speech/animal windows or one-second dropping windows are useful. This decomposition suggests that future zero-shot work should model query-conditioned temporal resolution before adding a more expressive residual scorer.

C. What remains to be confirmed

Cross-fitting prevents a held video’s own outcomes from selecting its candidate, and 9/10 class choices are fold-stable. It does not undo candidate design after benchmark exposure, class-label supervision, or reuse of the public validation split. A fresh natural dataset with an independently frozen router is required to test the diagnosis. Until then, 0.096995 is an in-domain exploratory estimate rather than a new zero-shot headline.

VII. THREATS TO VALIDITY

a) Construct validity.: Class-macro temporal AP rewards annotated overlap. It does not measure explanation, causal reasoning, source separation, or answer sufficiency. Queries are three templates for ten known classes, not unconstrained open-vocabulary language.

b) Internal validity.: The candidate pools reproduce all ten capped source files exactly. A first router was invalidated because its class join could violate the shared cap; no router report from that attempt is used. The corrected analyzer’s only post-freeze code amendment replaces a quadratic cap assertion with an equivalent linear count. Both records are retained.

c) Metric validity.: The frozen implementation differs slightly from the pinned upstream ActionFormer evaluator in stable tie ordering and floating-point tIoU association. Reproducing both upstream semantics matches upstream class AP within $5.55e-17$; the maximum frozen-versus-upstream mAP difference is 0.00001873, and every metric-dependent frozen gate remains unchanged.

d) External and statistical validity.: Perception Test contains ten everyday sound classes and short recordings. Results need not transfer to music, bioacoustics, meetings, long recordings, new languages, or strongly polyphonic scenes. Video bootstraps summarize this finite panel; classwise diagnostic criteria are not multiplicity-adjusted hypothesis tests.

VIII. REPRODUCIBILITY, RIGHTS, AND COMPUTE

The release includes protocols, freeze and amendment locks, QCR checkpoint hashes, exact candidate-cap reconstruction receipts, full and CLAP-only router predictions, natural predictions, aggregate reports, source data, tests, and independent verifiers. Raw benchmark audio, CLAP weights, official ActionFormer features/checkpoint, UrbanSound8K, ESC-50, and LibriSpeech remain with their providers. Exact URLs, revisions, checksums, and acquisition instructions are retained; non-commercial source restrictions are disclosed.

All natural embeddings use the same pinned CLAP encoder and verified precompute backend. The largest frozen QCR residual is below the analytic 0.25 bound. Cross-fit candidate

pools are excluded from the compact submission archive because they are regenerable derived intermediates; their row counts and SHA-256 values remain in the receipt.

IX. CONCLUSION

A bounded residual ranker can be genuinely effective for controlled exact-onset evidence retrieval yet fail fixed zero-shot localization in natural audio. Retaining that failure and enforcing the shared output cap reveals the main mechanism: class-dependent temporal scale, not the QCR residual, supplies most of the in-domain cross-fitted gain. QCR adds a smaller positive increment, but the diagnostic router remains post-outcome and below supervised ActionFormer; it also does not improve held-out answers. The result is a sharper, reproducible boundary on where query-conditioned ranking transfers and what must be confirmed next.

AUTHOR DECLARATIONS

Names, affiliations, ORCIDs, corresponding contact, CRediT roles, funding, competing interests, data-license review, and any required AI-assistance disclosure must be supplied and approved by the accountable authors before submission. No new human-subject data were collected; the study analyzes public benchmark releases under their stated terms.

REFERENCES

- [1] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, “CLAP: Learning audio concepts from natural language supervision,” in *Proc. IEEE ICASSP*, pp. 1–5, 2023.
- [2] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, “Large-scale contrastive language–audio pretraining with feature fusion and keyword-to-caption augmentation,” in *Proc. IEEE ICASSP*, pp. 1–5, 2023.
- [3] A. Guzhov, F. Raue, J. Hees, and A. Dengel, “AudioCLIP: Extending CLIP to image, text and audio,” in *Proc. IEEE ICASSP*, pp. 976–980, 2022.
- [4] H.-H. Wu, P. Seetharaman, K. Kumar, and J. P. Bello, “Wav2CLIP: Learning robust audio representations from CLIP,” in *Proc. IEEE ICASSP*, pp. 4563–4567, 2022.
- [5] X. Xu, H. Dinkel, M. Wu, and K. Yu, “Text-to-audio grounding: Building correspondence between captions and sound events,” in *Proc. IEEE ICASSP*, pp. 606–610, 2021.
- [6] H. Tang, Y. Wang, J. Zhu, S. Zhang, M. Xu, Q. Zheng, and Y. Hu, “Listen as you wish: Audio-based event detection via text-to-audio grounding in smart cities,” arXiv:2106.14136, 2021.
- [7] X. Xu, Z. Ma, M. Wu, and K. Yu, “Towards weakly supervised text-to-audio grounding,” arXiv:2401.02584, 2024.
- [8] Z. Zhang, X. Cheng, W. Yan, T. Zhang, D. Fu, B. Zhang, Y. He, and T. Jin, “Auto-AEG: Scalable data construction for open-vocabulary audio event grounding,” arXiv:2607.04383, 2026.
- [9] N. Turpault, R. Serizel, S. Wisdom, H. Erdogan, J. Hershey, E. Fonseca, P. Seetharaman, and J. Salamon, “Sound event detection and separation: A benchmark on DESED synthetic soundscapes,” in *Proc. IEEE ICASSP*, pp. 840–844, 2021.
- [10] C. Bilen, G. Ferroni, F. Tuveri, J. Azcarreta, and S. Krstulović, “A framework for the robust evaluation of sound event detection,” in *Proc. IEEE ICASSP*, pp. 61–65, 2020.
- [11] C. Zhang, J. Wu, and Y. Li, “ActionFormer: Localizing moments of actions with Transformers,” in *Proc. ECCV*, pp. 492–510, 2022.
- [12] V. Pătrăucean *et al.*, “Perception Test: A diagnostic benchmark for multimodal video models,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023.
- [13] J. Chalk, J. Huh, E. Kazakos, A. Zisserman, and D. Damen, “TIM: A time interval machine for audio-visual action recognition,” in *Proc. IEEE/CVF CVPR*, pp. 18153–18163, 2024.
- [14] N. Bodla, B. Singh, R. Chellappa, and L. S. Davis, “Soft-NMS—Improving object detection with one line of code,” in *Proc. IEEE ICCV*, pp. 5561–5569, 2017.